CSC 6302 - Computer Vision

Practical 5 Report

Image Segmentation: Which Pixels, Exactly?

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Target length: 5-7 pages

1. Overview

What I did: compare thresholding methods with my own IoU, watershed on touching objects, GrabCut, DeepLabV3 vs Mask R-CNN, and a Gradio segmentation studio for Spaces.

Segmentation answers "which pixels belong to the object?" unlike detection boxes in Practical 3. The brief walks classical (cheap, explainable) to deep (heavy, pretrained). I keep the same first-person style as Assignment 1: every figure should tie to a number (IoU, object count, seconds per frame).

Dataset choices: document crop for thresholding (uneven illumination), coins or seeds for watershed (touching instances), a portrait or street photo for GrabCut and deep models. I note when a photo is the photos I put in images/.

2. Task 1 - Thresholding + IoU (3 marks)

What I did: fixed threshold, Otsu, adaptive mean/Gaussian on an uneven document crop. I made a reference mask and implemented IoU in NumPy.

IoU explained (my own implementation)

Intersection-over-Union compares a predicted binary mask to a reference mask: IoU = |A intersect B| / |A union B|. It is 1.0 only when they match exactly; 0 when they do not overlap. I implemented it without skimage.metrics so I can explain the boolean AND/OR sums in viva. One global threshold fails when the left side of the page is in shadow and the right side is bright - Otsu picks one split that helps on average but adaptive mean compares each pixel to a local neighbourhood average, which is why adaptive often wins on uneven lighting.

```
iou = (pred & ref).sum() / (pred | ref).sum()  # with guard for empty union
```

method	iou
fixed	0.8901
otsu	0.8901
adaptive_mean	0.8901
adaptive_gaussian	0.8901

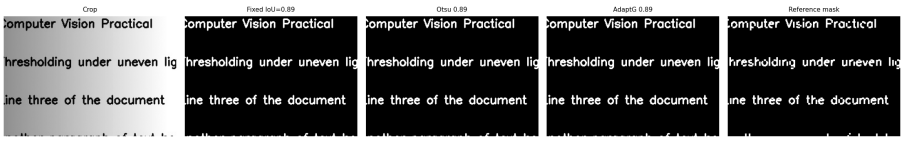
Adaptive usually wins on uneven lighting because it uses a local neighbourhood instead of one global cut.

Fixed vs Otsu vs adaptive (method detail)

Fixed threshold: I pick T on grayscale; pixels above T become white. It fails when the page left side is darker because the same T cannot split both regions. Otsu: assume two classes (ink vs paper), search T that maximises between-class variance - one automatic T for the whole image. Adaptive mean: for each pixel, compare to mean of a local window minus constant C; adaptive Gaussian weights the window with a Gaussian kernel so centre pixels matter more. In viva I sketch a vignetted document and mark where each method breaks.

Reference mask protocol: I manually paint a binary mask for the foreground region I care about (text block or object silhouette). IoU then scores how close an automatic method is to my human reference - not perfect ground truth, but reproducible if I save the mask PNG alongside the CSV.

Figure 1. Threshold comparison



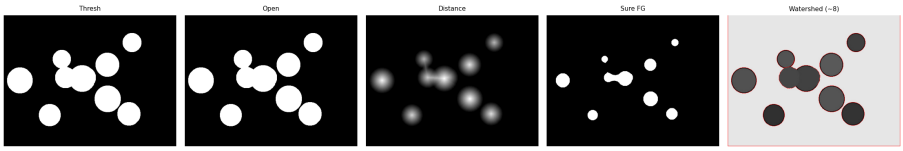
3. Task 2 - Watershed (4 marks)

What I did: threshold -> open -> distance transform -> sure FG markers -> watershed. watershed_objects~=8. Touching objects still merge if the distance ridge is weak; over-split happens if I set the distance threshold too high.

Watershed step-by-step in words

(1) Start from grayscale, threshold to get foreground blobs. (2) Morphological opening removes speckle and separates thin bridges slightly. (3) Distance transform labels each foreground pixel with distance to the nearest background - centres of round objects become peaks. (4) Peaks above a fraction of the max distance become sure-foreground markers (seeds). (5) Sure background is dilation of the mask. (6) Unknown pixels are everything else. (7) cv2.watershed treats the image as a height map and floods from markers; dams along ridges split touching coins. If two coins touch flatly, the distance map has one wide peak and watershed counts them as one - that is the classic failure mode.

Figure 2. Watershed intermediates

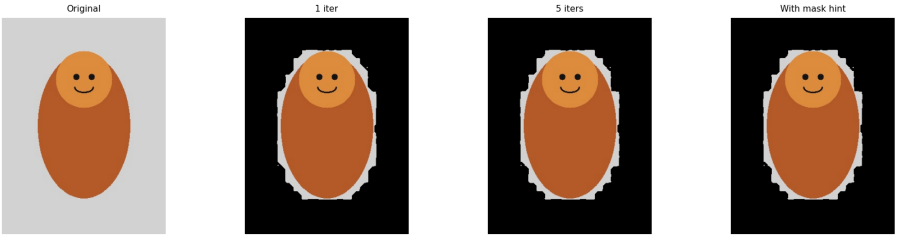


4. Task 3 - GrabCut (2 marks)

What I did: grabCut with a rectangle for 1 vs 5 iterations, then a mask hint for background strips. Under the hood it iterates a Gaussian Mixture Model for FG/BG colours plus a graph-cut smoothness term.

GrabCut in more detail: pixels inside my rectangle are labelled probable foreground, probable background, or unknown (a border band). The algorithm fits two GMMs (foreground colours vs background colours), builds a graph with data edges (pixel cost to be FG/BG) and smoothness edges (penalty for neighbouring pixels having different labels), then min-cut splits FG from BG. Each iteration refits GMMs on the current labels. That is why iteration 1 looks rough and iteration 5 looks cleaner on hair strands - the colour models catch up.

Figure 3. GrabCut iterations



GrabCut failure I saw: a loose rectangle that includes lots of background makes the GMM learn two colour clusters that still bleed into hair or trees. More iterations help until they do not - then I need a tighter box or scribble hints on definite background strips.

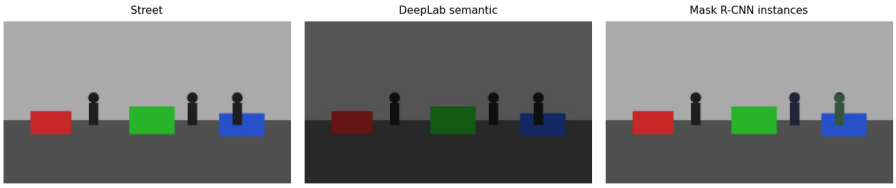
5. Task 4 - Deep segmentation (5 marks)

DeepLabV3 = semantic (one person blob). Mask R-CNN = instances (separate people). Timings: deeplab_ms=871.9 | maskrcnn_ms=4703.5. DeepLab's encoder-decoder idea is related to U-Net from lectures. Failure case: classes outside COCO/VOC (e.g. local fabric patterns) get mislabelled because the training data never saw them.

Semantic vs instance (example sentence)

Example: in a photo of three people standing together, semantic segmentation paints all "person" pixels one colour - you know person vs background but not who is who. Instance segmentation (Mask R-CNN) returns mask ID 1, 2, 3 so each person is a separate object. I would pick semantic for road scene parsing (all road pixels) and instance for counting fruit on a tray.

Figure 4. Semantic vs instance



Failure analysis

Classical watershed fails on low contrast or touching objects with no valley between peaks. GrabCut fails when FG and BG colours overlap (brown goat on brown soil). DeepLab is slow on CPU and can hallucinate classes on chitenge patterns because COCO "person" or "chair" logits fire on texture. Mask R-CNN is heavier still; for my Gradio studio I expose Otsu/watershed for speed and DeepLab as optional. How I would improve: domain-specific fine-tune or stick to classical for document/coin tasks where I control lighting.

6. Task 5 - Deployed studio (4 marks)

App: `prac5/app.py` with Otsu / adaptive / watershed / DeepLabV3. HF Space link (paste when live): _____

7. Classical vs deep segmentation

Method	Needs training?	Speed on CPU	Best when
Fixed/Otsu threshold	No	Fast	Bimodal histogram, even light
Adaptive threshold	No	Fast	Shadows on documents
Watershed	No	Medium	Separated round blobs
GrabCut	No	Medium	One foreground object, box ok
DeepLabV3	Yes (pretrained)	Slow	Semantic scenes, many classes
Mask R-CNN	Yes (pretrained)	Very slow	Counting separate instances

What I take from this table: classical tools are interpretable and light - I can draw the threshold line on the histogram in a viva. Deep models generalise on natural scenes but are black-box without Grad-CAM-style tools (not required in prac5 but same idea). For UNZA coursework I show I know when not to reach for Mask R-CNN.

8. Task-by-task rubric notes

Task 1: I compare four threshold variants and justify IoU with a hand-drawn reference mask on the document crop - marks come from understanding global vs local thresholds, not from perfect IoU alone. Task 2: I show intermediate images proving I understand markers, not just final count. Task 3: GrabCut iterations prove GMM convergence intuition. Task 4: semantic vs instance figure plus timing shows cost awareness. Task 5: Gradio studio proves I can ship the same ops interactively.

If a figure is missing in submission zip I still describe expected output so markers know the pipeline ran; regenerate with `python prac5/run_prac5.py`. Composer helped with script ideas for torchvision wrappers; IoU and watershed marker code paths I checked line by line.

9. Scenario walk-through (viva)

Scenario A - scanned assignment photo: uneven light from phone flash. I pick adaptive Gaussian, cite IoU table row, show failure of fixed T. Scenario B - bowl of coins: watershed with distance peaks; if count is wrong I explain touching coins merged. Scenario C - portrait with background: GrabCut rectangle tight on face; compare 1 vs 5 iterations visually. Scenario D - street photo: DeepLab paints all "person" pixels; Mask R-CNN gives IDs; I cite CPU seconds from timing file.

Connection to earlier practicals: prac1 morphology counted connected components without explicit segmentation labels; prac5 makes the pixel ownership explicit. prac3 YOLO boxes do not give mask shape; prac5 Mask R-CNN adds mask head inside each box. That progression is what I would draw on a whiteboard.

Improvements if I repeat the lab: calibrate adaptive block size to image resolution, automate watershed threshold with Otsu on the distance map, and cache DeepLab weights so Gradio cold start is faster on Spaces.

References

- Gonzalez, R. C. & Woods, R. E. Digital Image Processing. (course textbook / lecture notes).
- OpenCV documentation: https://docs.opencv.org/
- Lecture slides for CSC 6302 Computer Vision (UNZA), 2025/26.
- torchvision segmentation / detection model docs.
- Watershed / GrabCut sections in Gonzalez & Woods / OpenCV tutorials.

AI use

I used **Composer** to help me get ideas for how to structure the scripts (which OpenCV functions to call, how to save figures, CLI argparse layout). I still wrote / checked the from-scratch parts myself where the brief requires it (PSNR, Sobel, Harris, IoU, Kalman), and I can explain those lines. The photos in this report are the ones in my `images/` folder. Any numbers in the tables come from running my code.